# Unit 3 – FMEA, Taguchi, QFD, TPM and Quality Systems: Priority-Ordered Exam Notes

19MEE347 Total Quality Management · Built from your 8 Unit 3 PPTs (FMEA, QLF, QFD, TPM, ISO 9000:2000, Quality Audit, ISO 14000, 5S)

## How to use these notes

Topics follow the priority list from my exam prediction, ordered by study value inside each section. Importance tags:

- 🔴 **VERY IMPORTANT** – learn fully and practise writing/solving
- 🟡 **MODERATELY IMPORTANT** – key points plus one example
- 🟢 **JUST READ** – skim once

Your teacher said Unit 3 carries **20–25 marks** and numericals total **25–30 marks** of the paper, and told the class to study FMEA and practise its numericals. Everything else (percentages, mark weights, question forms) is **my estimate**, not a guaranteed question. I expect roughly 8–12 numerical marks inside Unit 3 (Taguchi and FMEA mainly).

### ⚠️ Check these slide issues first

1. **Taguchi slide 17:** it says L(0.10) = 125,000 × (0.10)² = $12.50. That arithmetic gives **$1,250**. The $12.50 figure is correct for a deviation of **0.01**. Use 0.01 if you quote that example.
2. **FMEA slides 19 and 21** describe the scales as “0 to 10”, but the rating tables and slide 29 use **1 to 10**, and the RPN range is **1 to 1000**. Write 1–10.
3. **ISO 9001:2000 structure:** the slide says “21 elements in five sections”, but the items listed under the five sections add up to 23. Write “five major sections” and list the items.
4. ISO 9000:2000 is the version in your syllabus (it was revised later). Answer as per your slides.
5. **Not in your slides:** QS 9000, and “need for quality systems” as a separate topic. I added short notes (marked standard).

# Priority map

| # | Topic | Importance | Type | Est. weight in Unit 3 | Share of your Unit 3 time |
|---|---|---|---|---|---|
| A1 | FMEA | 🔴 VERY IMPORTANT (~90%) | Numerical + theory | 7 marks | 20% |
| A2 | Taguchi quality loss function | 🔴 VERY IMPORTANT (~90%) | Numerical + theory | 6 marks | 20% |
| A3 | QFD and House of Quality | 🔴 VERY IMPORTANT (~75%) | Theory (weighted score possible) | 4 marks | 15% |
| A4 | TPM (pillars, losses, OEE) | 🔴 VERY IMPORTANT (~70%) | Theory + OEE numerical | 3–5 marks | 14% |
| B1 | ISO 9000:2000 | 🟡 MODERATE (~60%) | Theory | 3–5 marks | 9% |
| B2 | Quality audit | 🟡 MODERATE (~50%) | Theory | 3–5 marks | 7% |
| B3 | ISO 14000 (and QS 9000) | 🟡 MODERATE (~50%) | Theory | 3–4 marks | 6% |
| B4 | 5S | 🟡 MODERATE (promoted from Just Read) | Theory | short note | 5% |
| C1 | Need for quality systems, documentation, implementation | 🟢 JUST READ (~40%) | Theory | part of ISO answers | 4% |

ISO 9000, audit, ISO 14000 and 5S compete for roughly 5 marks, so expect only one or two of them. I moved **5S** up from Just Read because your teacher's Unit 3 set includes its own 5S deck and TPM sits on a 5S foundation. It is a quick-win topic.

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# SECTION A – 🔴 VERY IMPORTANT

## A1. FMEA – Failure Mode and Effects Analysis 🔴 (named by your teacher)

**Definition:** a methodology to evaluate failure modes and their effects in designs and processes. It is a structured approach to identify how a product or process can fail, estimate the risk of specific causes, prioritise actions to reduce risk, and evaluate the validation or control plan.

**Failure mode:** the way in which a component, subassembly, product, input or process could fail to perform its intended function. Failure modes can be caused by upstream operations or can cause downstream operations to fail. “Hidden” examples: cracking, electrical short circuit, torque fatigue, excessive deformation.

**History:** first used in the 1960s in aerospace (Apollo missions); the US Navy issued MIL-STD-1629 in 1974; in the late 1970s the automotive industry adopted it because of liability costs.

**Why use it:** facilitates process improvement; finds and removes concerns early in a design or process; improves internal and external customer satisfaction; focuses on **prevention**; may be a customer or quality-system requirement. It shows which areas affect customers most and which failures are hardest to detect.

**Types: Design FMEA** (analyses product design before release, with focus on function; also systems and subsystems at concept stage) and **Process FMEA** (analyses manufacturing and assembly processes).

**When to conduct:** early in a process improvement study; when new systems, products or processes are designed; when existing designs or processes change; when carry-over designs are used in new applications; after functions are defined but before hardware is selected or released.

**Team:** led by a Black Belt or responsible manufacturing/technical person familiar with FMEA. Members: design engineers, process engineers, materials/suppliers, operators, reliability, customers.

## Procedure (as in your slides)

1. For each process input, find how it can go wrong: the **failure mode**.
2. For each failure mode, find the **effects** and assign **Severity (S)**.
3. Identify the **causes** and assign **Occurrence (O)**.
4. List the **current controls** and assign **Detection (D)**.
5. Calculate the **Risk Priority Number (RPN)**.
6. Recommend actions, assign responsible persons, take action. Give priority to high RPNs, and **always look at any severity rated 10**.
7. Re-rate S, O and D after the action and compare the new RPN.

The flow diagram in your slides reads: identify failure mode → severity → occurrence → detection → RPN.

```
RPN = Severity (S) × Occurrence (O) × Detection (D)      range 1 to 1000
Slide example: S = 5, O = 2, D = 8  →  RPN = 5 × 2 × 8 = 80
```

## Rating scales (1–10 is the most common; 1–5 is easier to score but less precise)

| Rating | Severity (effect on customer) | Occurrence | Detection |
|---|---|---|---|
| 10 | Hazardous without warning | Almost inevitable | Absolute uncertainty (cannot detect) |
| 9 | Hazardous with warning | Very high | Very remote chance |
| 8 | Very high: system inoperable, destructive failure | High | Remote |
| 7 | High: inoperable with equipment damage | High | Very low |
| 6 | Moderate: inoperable with minor damage | Moderate | Low |
| 5 | Low: inoperable without damage | Moderate | Moderate |
| 4 | Very low: significant performance loss | Low | Moderately high |
| 3 | Minor: some performance loss | Low | High |
| 2 | Very minor | Very low | Very high |
| 1 | None | Nearly impossible | Almost certain to detect |

Severity is the importance of the effect on customer requirements (often you cannot change it); occurrence is how often a cause creates a failure mode; detection is the ability of current controls to find or prevent the cause. **For detection, a high number is bad** (hard to detect).

**FMEA form columns:** process step/input; potential failure mode; potential effects; severity; potential causes; occurrence; current controls; detection; RPN; recommended actions; responsibility; actions taken; revised S, O, D and RPN. The form groups into four tasks: identify failure modes and effects; identify causes and controls; prioritise; determine and assess actions.

**Inputs:** brainstorming, C&E matrix, process map, process history, procedures, knowledge, experience. **Outputs:** list of actions to prevent causes or detect failure modes, and a history of actions taken. Failure modes and effects are not always one-to-one.

### Numerical practice (my own example)

| Failure mode | S | O | D | RPN | Rank |
|---|---|---|---|---|---|
| Bearing seat oversize | 8 | 3 | 6 | 144 | 1 |
| Surface scratch | 3 | 7 | 4 | 84 | 2 |
| Missing keyway | 9 | 2 | 3 | 54 | 3 |

Action on the top item: fit an in-process gauge, so D drops from 6 to 2. New RPN = 8 × 3 × 2 = **48** (down from 144). Comment: although the missing keyway ranks third, its severity is 9, so it still needs attention.

**Answer template (7 marks):** definition (2 lines) → types → 7 procedure steps → scales and RPN formula → one RPN example → benefits.

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## A2. Taguchi quality loss function (QLF) 🔴 (numerical)

**Why it matters:** the whole of Assignment 2 is about it, and the CDP gives it two lectures.

**Background:** introduced by Dr Genichi Taguchi (1980). Unique aspects: his definition of quality, the quality loss function, and robust design. It combines **cost, target and variation**; specification limits are of secondary importance.

**Taguchi's definition of quality:** the loss imparted to society from the time the product is shipped. Societal loss includes failure to meet customer requirements or ideal performance, harmful side effects, and losses from producing unusable products or toxic by-products.

**Sony USA vs Sony Japan:** customers preferred Sony-Japan sets. Sony-USA sets were spread uniformly **inside** the tolerance (m ± 5); Sony-Japan sets followed a normal curve centred on target, with about 0.3% outside tolerance. Lesson: aim at the **target** and reduce variance, not just stay inside the limits.

```
Nominal-the-best:   L(y) = k (y − T)²
   L(y) = loss at y;  y = measured value;  T = target;  k = quality-loss
coefficient
   k = loss at tolerance limit ÷ (tolerance)²
Average loss:       L = k [ σ² + (mean − T)² ]          (σ = standard
deviation)
Break-even deviation for a fix costing C:   |y − T| = √(C / k)
```

The loss is zero at the target and rises with the **square** of the deviation, so even in-specification parts carry loss. Standard forms for the other two types (named in your slides, formulas not given): smaller-the-better L = k y²; larger-the-better L = k ÷ y².

**Worked problem from your slides (corrected):** spec 0.500 ± 0.020 cm; repair costs $50 at the limit.

- 50 = k (0.02)², so **k = 125,000**, and L(y) = 125,000 (y − T)².
- For deviation 0.01: L = 125,000 × 0.0001 = **$12.50** per part (the slide writes 0.10 by mistake).
- A $2 factory adjustment is worth doing when 2 = 125,000 (x − T)², so x − T = **0.004**. Beyond 0.004 from target it is cheaper to adjust, so the economic spec is **0.500 ± 0.004**.

## Assignment 2 answers (re-calculated)

| Q | Working | Answer |
|---|---|---|
| 1 | k = 50 ÷ 5² = 2; y = 103, T = 100 | k = 2; loss = 2 × 3² = **₹18** |
| 2 | Rod 1 (102): 2×4; Rod 2 (98): 2×4; Rod 3 (101): 2×1 | **₹8, ₹8, ₹2**; Rod 3 is lowest, ₹6 less than the others |
| 3 | k = 72 ÷ 6² = 2; deviation 6 mm | Loss ₹72 against ₹5 fix: **justified**, saving ₹67 (break-even deviation √(5/2) = 1.58 mm) |
| 4 | k = 25 ÷ 2² = 6.25 | A: 6.25×2² = **₹25**; B: 6.25×3² = **₹56.25**; choose Process A |
| 9 | k = 5, deviation 4 mm | 5×16 = **₹80** per part; 500 parts = **₹40,000** |

Your PDF skips Q5–Q8, so ask a classmate for them; they are probably variants of these.

**Average-loss example (my own):** k = 2, σ = 1, mean = 101, T = 100 → L = 2 [1 + 1] = 4 per part.

## Robust design (theory part)

- **Noise** = events that push performance away from target. Three kinds: **external** (usage conditions: temperature, vibration, humidity), **deterioration** (wear, material ageing, rust) and **unit-to-unit/internal** (piece-to-piece variation).
- Robust design makes the process or product insensitive to noise. Characteristics: **larger-the-better** (strength, mileage), **nominal-the-best** (dimensions) and **smaller-the-better** (defects, cost, time).
- **Three steps:** (1) **concept design** – choose the technology, prototype works under ideal conditions; (2) **parameter design** – choose control factors and their optimal levels by experiment (Taguchi's main focus); (3) **tolerance design** – set specification limits after parameter design; may raise cost (better materials).

- Taguchi DOE is a structured way to find the best combination of inputs.

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# A3. Quality function deployment (QFD) and the House of Quality 🔴

- **Origin:** introduced in Japan in 1966 by Yoji Akao. **Definition:** a systematic, organised approach to taking customer needs into account when designing new products and services or improving existing ones. It weaves user needs through the whole development cycle including manufacturing.
- **Voice of the customer (VoC):** the driving force. Customers dictate attributes; expectations may be vague, and must be taken literally rather than translated into what the organisation wants.
- **Types of customer information:** solicited/measurable/routine (surveys, trade trials); unsolicited/measurable (complaints, lawsuits); solicited/subjective (focus groups); solicited/haphazard (customer visits, consultants); unsolicited/haphazard (conventions, vendors, suppliers).

**Four phases (customer-driven product development):**

| Phase | Name | Converts | Into |
|---|---|---|---|
| I | Product planning | Customer voice | Design requirements |
| II | Part deployment | Design requirements | Parts requirements |
| III | Process planning | Parts requirements | Process requirements |
| IV | Production planning | Process requirements | Production standards |

**Benefits:** customer-driven (prioritises resources, uses competitive information); reduces implementation time (fewer mid-stream design changes, fewer post-launch problems); promotes teamwork (consensus, better communication at interfaces); provides documentation (rationale for design, a living document). Summary: shorter development cycle, lower start-up cost, fewer engineering changes, fewer oversights.

## House of Quality (HOQ) – parts to label

| Part | Meaning |
|---|---|
| Customer requirements (WHATs) | Voice of the customer; primary, secondary, tertiary levels |

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| Part | Meaning |
|---|---|
| Technical descriptors (HOWs) | Voice of the organisation |
| Relationship matrix (WHATs vs HOWs) | Strength: **9 strong, 3 medium, 1 weak** |
| Correlation matrix, the "roof" (HOWs vs HOWs) | **+9 strong positive, +3 positive, −3 negative, −9 strong negative** |
| Customer competitive assessment | Our product vs competitors A and B, rated 1–5 |
| Technical competitive assessment | Same comparison on technical measures |
| Prioritised customer requirements | Importance rating, target value, scale-up factor, sales point, absolute weight |
| Prioritised technical descriptors | Degree of technical difficulty, target value, absolute weight and %, relative weight and % |

**Steps to build:** (1) list customer requirements, (2) list technical descriptors, (3) fill the relationship matrix, (4) fill the correlation matrix, (5) competitive assessments, (6) prioritise customer requirements, (7) prioritise technical descriptors.

**Weighted-score numerical (standard method; check your class notes):**

```
Technical importance of a HOW = Σ (customer importance × relationship value, 9/3/1)
Absolute weight of a WHAT      = importance × scale-up factor × sales point
```

My example: WHAT importance 5, 3, 2. HOW 1 relationships 9, 3, 0 → 45 + 9 + 0 = **54**. HOW 2: 1, 9, 3 → 5 + 27 + 6 = **38**. HOW 3: 3, 0, 9 → 15 + 0 + 18 = **33**. Total 125, so relative weights are 43%, 30% and 26%; HOW 1 gets top priority.

Your slide 28 has a worked helmet-company HOQ as a picture; look at it once.

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## A4. Total Productive Maintenance (TPM) 🔴

- **Idea:** a holistic approach to equipment maintenance aiming at **perfect production: no breakdowns, no small stops or slow running, no defects, no accidents**. Operators help maintain their own equipment, with proactive and preventive maintenance.
- **Total** = everyone (maintenance and production) is involved; **Productive** = production goes on while actions are carried out; **Maintenance** = keep equipment as good as or better than original condition (keep in good condition, repair, clean, lubricate).
- **Evolution:** breakdown maintenance → preventive maintenance → productive maintenance → TPM.

- **Why (organisation):** higher OEE, less firefighting, lower operating cost, better delivery, more competitive. **Why (employees):** less pressure, better cooperation across departments, fewer accidents, higher job satisfaction and security.
- **Principles:** start with 5S/visual factory; train for operations and maintenance skills; involve everyone with cross-functional teams; use OEE as a compass; improve existing planned maintenance; reduce life-cycle cost; eliminate the six major losses aiming at zero unscheduled downtime.
- **Strategies:** planned maintenance system, initial control system, operator autonomous maintenance, loss elimination, education and training, and **three zeros** (defects, accidents, failures).
- **Measures:** short term – set up an autonomous maintenance programme and raise OEE by a target %; long term – reach plant OEE of **85% within 3 years**.

**TPM vs TQM:**

| Category | TQM | TPM |
|---|---|---|
| Object | Quality (output and effects) | Equipment (input and cause) |
| Orientation | Software oriented | Hardware oriented |
| Target | Quality for PPM | Elimination of losses and wastes |

Similarities: empowered employees, continuous improvement, senior management commitment, long-term outlook. TPM vs 6S: 6S improves worker productivity and organisation/visual indicators; TPM improves equipment effectiveness and maintenance–production partnership; both cover initial cleaning/repair and safety.

## Eight pillars of TPM (foundation: 5S)

| # | Pillar | Aim |
|---|---|---|
| 1 | Autonomous maintenance | Operators own routine cleaning, lubricating and inspection; frees maintenance staff |
| 2 | Focused improvement | Cross-functional teams remove recurring problems step by step |
| 3 | Planned maintenance | Schedule tasks by predicted or measured failure rates; cut unplanned stops |
| 4 | Quality maintenance | Prevent defects at source with root cause analysis; zero-defect conditions |

| # | Pillar | Aim |
|---|---|---|
| 5 | Early equipment management | Use TPM experience to design new equipment; faster, smoother start-up |
| 6 | Training and education | Fill skill gaps for operators, maintenance staff and managers |
| 7 | Safety, health and environment | Safe, accident-free workplace |
| 8 | Administration (office) TPM | Apply TPM to admin functions; remove office losses |

Memory hook: **Auto, Focus, Plan, Quality, Early, Train, Safety, Office.**

## Six major losses

| Loss | OEE category | Examples |
|---|---|---|
| 1 Unplanned stops (equipment failure) | Availability | Tool failure, motor failure |
| 2 Setup and adjustments | Availability | Changeover, material shortage, warm-up (reduced by SMED) |
| 3 Small stops (idling, minor stoppages) | Performance | Jams, blocked sensor; usually under 5 minutes |
| 4 Slow running (reduced speed) | Performance | Incorrect setting, wear, misalignment |
| 5 Production defects | Quality | Scrap, rework in steady state |
| 6 Reduced yield | Quality | Rejects during start-up or warm-up |

Grouped as **downtime losses** (1, 2), **speed losses** (3, 4) and **defect losses** (5, 6).

## OEE (Overall Equipment Effectiveness)

```
OEE = Availability × Performance × Quality
Availability = run time ÷ planned production time
Performance  = (ideal cycle time × total count) ÷ run time
Quality      = good count ÷ total count
```

Benchmarks (your slide): 100% is perfect production; **85% is world class** for discrete manufacturers; 60% is typical; 40% is not unusual without TPM or lean. The slide gives the three components and goals (availability: no stops; performance: no small stops or slow running; quality: no defects); the formulas above are the standard ones.

Example (mine): planned 480 min, downtime 47 min → run time 433; availability = 90.2%. Ideal cycle 1 min/unit, 400 units made → performance = 400 ÷ 433 = 92.4%. 392 good units → quality = 98%. **OEE = 0.902 × 0.924 × 0.98 ≈ 81.7%**, below the 85% world-class level.

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# SECTION B – 🟡 MODERATELY IMPORTANT

## B1. ISO 9000:2000 🟡

- **Overview:** quality system standards adopted by the International Organization for Standardization in 1987, revised in 1994 and 2000. They are technical specifications and criteria used as rules or guidelines so materials, products, processes and services are fit for purpose.
- **Objectives:** achieve, maintain and improve product quality; improve operations to continually meet customer and stakeholder needs; give confidence that quality requirements are fulfilled.
- **Key characteristics:** establishes a quality management system (QMS) for consistency; **not prescriptive** (specifies "what", not "how"); **not a product standard; not TQM; site specific**.
- **The three standards:** ISO 9000:2000 (fundamentals and vocabulary), **ISO 9001:2000** (requirements), ISO 9004:2000 (guidelines for performance improvement).
- **Eight quality management principles:** customer focus, leadership, involvement of people, process approach, system approach to management, continual improvement, factual approach to decision making, mutually beneficial supplier relationships.

**Structure of ISO 9001:2000 – five sections:**

| Section | Items |
|---|---|
| System requirements | Establish the QMS; document the QMS |
| Management responsibility | Management commitment, customer focus, quality policy, planning, administration, management review |

| Section | Items |
|---|---|
| Resource management | Provision of resources, human resources, facilities, work environment |
| Product/service realisation | Planning of realisation, customer-related processes, design and development, purchasing, production and service operations, control of measuring and monitoring devices |
| Measurement, analysis and improvement | Planning, measurement and monitoring, control of nonconformity, analysis of data, improvement |

Memory hook: **S-M-R-P-M** (System, Management, Resources, Product, Measurement).

**Documentation levels:** manual (philosophy, policy, objectives, approach) → procedures → work instructions (department, product, process) → records (proof and objective evidence) → ad-hoc temporary documents.

**Certification process:** (1) implement the control and documentation procedures; (2) thorough audit by an independent registrar licensed by an accreditation body; (3) on compliance, receive a certificate and get listed in a directory of registered suppliers; (4) periodic surveillance, with full audits every few years.

**Benefits of registration:** documented QMS, reduced variation, business growth, fewer customer audits, lower cost and higher profit, better communication, greater quality awareness, training for all, competitiveness, no duplicated quality systems.

**Problems:** cost and time (application and maintenance), level of internal expertise, executive commitment, choice of registrar. Your slide notes that some began to question its usefulness.

## B2. Quality audit 🟡

- **Definition:** a systematic and independent examination to decide whether quality activities and results comply with planned arrangements, whether they are implemented effectively, and whether they suit the objectives. **Goal:** collect objective evidence to make an informed judgement about the system or product audited.
- **Types by who audits: internal (first party/self)** – employees, consultants or contractors; **second party (supplier audit)** – a customer audits you, or you audit your supplier; **third party** – an independent organisation.
- **Sub-types:** compliance (do we comply with the standard; desk audit), **system** (the theory, e.g. document control), **process** (the practice, e.g. manufacturing), **product** (the result, finished goods against specifications).

- **Terms:** auditor (qualified person who audits), client (requests the audit; for internal audits it is the management representative), auditee (audited party), observation (fact backed by objective evidence), nonconformity (non-fulfilment of a specified requirement), objective evidence (verifiable information from observation, measurement or test), quality system.
- **Reasons – internal:** find the level of compliance, build confidence and interdepartmental trust, find improvements (premises, equipment, operations, personnel and training), stimulate improvement, recommend corrective action, monitor improvement. **External:** check a supplier's capability to deliver fit-for-purpose goods and services, build mutual confidence and communication.
- **Seven steps (your cycle diagram):** plan and prepare → arrange and announce → go to site, meet and explain → perform the audit → informal oral report of findings → formal report with recommendations → follow-up.
- **Process audit:** examines established methods, instructions, work flow, equipment maintenance programmes, material handling and housekeeping. It validates procedures; processes are parts of a system and interact, so communication between them matters.
- **Auditor's tools:** documents (standards, previous audit reports, the auditee's records, checklists), the auditor's own eyes, ears and judgement, and the audit plan.

## B3. ISO 14000 (and QS 9000) 🟡

- **What:** the first internationally accepted series of **environmental management system (EMS)** standards; voluntary; a guide to principles, systems and supporting techniques. Aim: support environmental protection and pollution prevention in harmony with socio-economic needs.
- **EMS:** a way of doing business while managing environmental risks; adds order and consistency; builds environmentally responsible practice into daily processes; gives measurable improvement.
- **Why implement:** better compliance with environmental law, sustainable development, credibility, pressure from environmentalists, worldwide environmental focus.
- **Standards in the series: 14001** = EMS requirements (companion 14004); **14010 series** = auditing; **14020** = environmental labelling; **14030** = environmental performance evaluation; **14040 series** = life cycle assessment (LCA).
- **Certification process (6 steps):** preliminary assessment → document review → initial assessment → main assessment → certification/registration → surveillance.
- **Company examples in your slides:** L&T, Ashok Leyland, BILT, Toyota Kirloskar and Nippon Toyota (policy pictures for Ashok Leyland and Toyota Kirloskar).

- **QS 9000 (not in your slides; standard note):** the automotive supplier quality standard created by US car makers on top of ISO 9001 (adds customer-specific and continuous-improvement requirements); it has since been replaced by IATF 16949. Check your class notes for what your teacher expects.

## B4. 5S 🟡 (promoted; easy marks)

| Japanese | English | Meaning |
|---|---|---|
| Seiri | Sorting | Remove unnecessary tools, parts and instructions; keep only essentials in accessible places |
| Seiton | Setting in order / straightening | A place for everything and everything in its place; label locations; most-used items nearest |
| Seiso | Shining / systematic cleaning | Clean workspace and equipment daily and keep it tidy |
| Seiketsu | Standardising | Identical work stations and tool locations; everyone knows their responsibilities for the first three S |
| Shitsuke | Sustaining / self-discipline | Maintain and review standards; don't slide back; look for better ways |

**Benefits:** better efficiency and speed and quality of work, less waste, less employee frustration, better safety, visually attractive workplace, lower direct and hidden costs, higher motivation. **Objectives:** productivity, safety, reduced waste, worker commitment.

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# SECTION C – 🟢 JUST READ

## C1. Need for quality systems, documentation and implementation

- **Need (from the ISO 9000 objectives):** consistency, confidence for customers and management, and a base for continual improvement. Standard addition: customers and contracts often require a certified system.
- **Documentation:** the five-level hierarchy listed in B1 (manual, procedures, work instructions, records, ad-hoc documents).
- **Implementation:** put control and documentation procedures in place, train staff, run internal audits, then go through registrar certification and surveillance (steps in B1 and B2).

# Likely question skeletons (estimates, not guaranteed)

| Likely question | Marks | Fast answer plan |
|---|---|---|
| Taguchi: find k, compute loss, compare, decide on adjustment | 5–6 | A2 formulas and the five assignment answers |
| FMEA: calculate RPN, rank, recommend action | 5–7 | A1 table, procedure and the severity-10 rule |
| Explain FMEA procedure and types | 4–7 | A1 definition, 7 steps, design vs process |
| QFD and House of Quality (draw and explain) | 4–8 | A3 phases table and HOQ parts |
| TPM pillars, six losses, OEE (numerical) | 3–6 | A4 pillars table, losses table, OEE example |
| ISO 9000:2000 structure and certification | 3–5 | B1 five sections, documentation levels |
| Quality audit types and steps | 3–5 | B2 |
| ISO 14000 / 5S short note | 2–4 | B3, B4 |

## 40-minute last-minute plan

1. 10 min – Taguchi: redo Assignment 2 Q1–4 and Q9 without looking
2. 10 min – FMEA: scales, RPN formula, two ranking problems, then the 7 steps
3. 7 min – QFD: 4 phases, HOQ parts, one weighted-score calculation
4. 7 min – TPM: 8 pillars hook, six losses table, one OEE calculation
5. 6 min – ISO 9000 five sections, audit steps, 5S names
6. Skim ISO 14000 standards list if time remains

## Notes on source and limits

- All content comes from your 8 Unit 3 PPTs. TPM and several FMEA, QFD and audit slides are pictures; I read them as images. I did not study the helmet-company HOQ picture (QFD slide 28) in detail.

- Items marked “my own example”, “standard” or “not in your slides” are my additions, so check them against your class notes.
- Importance levels and mark weights are my estimates. Only the 20–25 and 25–30 mark hints and the FMEA message come from your teacher.